

Safety Evaluation of Responsibility-Sensitive Safety (RSS) on Autonomous Car-Following Maneuvers Based on Surrogate Safety Measurements

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Abstract— This paper evaluates the safety effect of Responsibility-Sensitive Safety (RSS) model on autonomous car-following maneuvers. The RSS model is embedded into Model Predictive Control (MPC) based Adaptive Cruise Control (ACC) algorithm. Car-following scenarios with sudden deceleration of lead vehicle at various front gap and velocity conditions are simulated. Vehicle movements characteristics and surrogate safety measurements are analyzed to evaluate safety performance of simulation before and after RSS is embedded. Results show that ACC safety performance can be improved significantly by RSS at large front gap conditions. As ACC is an optimization-based algorithm, large initial front gap leads to late deceleration decision and thus result in low safety performance. RSS model, which is derived from drivers' perception and reaction according to emergencies, can be applied as a safety guarantee to improve safety performance of such optimization-based algorithms.

I. INTRODUCTION

Ensuring the safety of Autonomous Vehicles (AV) functions is an inevitable challenge on the way to fully autonomous driving. According to the latest autonomous vehicle disengagement reports (vehicle gives control back to driver) released by California's Department of Motor Vehicles (DMV) in 2019, Waymo drove 1.2 million miles in California in 2018 and the disengagement rate dropped to 0.09 per 1,000 self-driven miles [1]. Although this rate ranks the best among all companies, the vehicles being tested are not ready to assure on-road safety. Moreover, as the AV industry involves multiple stakeholders, it is crucial to have a standardized and verified safety assurance for the operation of AV.

The Responsibility-Sensitive Safety (RSS) model, developed by Intel Mobileye, is a mathematical model for safety assurance of AV [2]. RSS ensures that the AV will stay in a "safe state" and not be responsible for an accident by formalizing "common sense" rules and human judgement of what it means to drive safely. RSS ensures that AVs will not be responsible for causing accidents. Therefore, RSS can be applied as a constraint to AV control algorithms to enhance safety. The RSS model defines safe distances for all driving scenarios and the proper response to be taken in order to evade a dangerous situation. Furthermore, RSS creates a set of parameters to be calibrated to make AVs safer and more

"human-like". Different combinations of parameters have a significant impact on the traffic flow. It is found that the two deceleration parameters of the model as well as the vehicle-controller's response time are the significant factors, and the maximum deceleration influences the result but not as much as others [3].

Previous studies have approved that active safety functions, such as Adaptive Cruise Control (ACC), which is already commercialized on the automobile market, can improve traffic flow, driving comfort and reduce traffic accidents [4,5]. However, as the goal of ACC is to maintain a safety distance, if the lead vehicle suddenly decelerates or if another vehicle suddenly cuts-in in front of the subject vehicle, the ACC is deactivated, and driver becomes responsible for breaking [6,7]. As the goal of AV function development is fully automation, an improved longitude control framework which can response safely under extreme conditions is required.

In this paper, a Model Predictive Control (MPC) based ACC algorithm is modified by embedding with the RSS safety assurance model [8]. Two microscopic car-following simulation scenarios are designed to evaluate the safety performance of ACC with RSS according to National Highway Traffic Safety Administration (NHTSA) pre-crash scenarios [9]. Safety performance of ACC algorithm with and without RSS is estimated through computing well-developed surrogate safety indicators. The paper aims to evaluate safety impact of RSS on both regular and extreme car-following conditions. Furthermore, this paper aims to suggest proper RSS applications and parameter sets to achieve safety assurance on both AV and surrounding human-driven vehicles.

II. SURROGATE SAFETY MEASUREMENTS (SSM)

Traffic Conflict Technique (TCT) is a safety assessment approach based on analyzing vehicle dynamics. As potential conflicts may lead to slight, and near crashes, conflict analysis based on TCT provides an alternative approach to evaluate safety performance without enough crash data [10,11]. Surrogate safety measurements (SSM), also known as safety indicators, are commonly used to quantify occurrences and severity of potential risks involving vehicle conflicts [12]. Such measurements are derived from vehicle interaction analysis,

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including time-dependent vehicle movements, risk avoidance behavior, or safety margin of conflicts [13].

Time to Collision (TTC) is the most widely used SSM [12]. It is defined by the time to a potential collision between two vehicles, as shown in Figure 1a [14]. Therefore, a smaller TTC represents a severer conflict. Generally, risks are identified for any two vehicles with a TTC value less than a given threshold, which is normally chosen as 3s. Minimal TTC value is commonly used to evaluate conflict severity of a dynamic conflict process. A severe conflict is normally identified by a minimal TTC of less than 1s. TTC is computed by the following equation.

$$TTC_i(t) = \begin{cases} \frac{x_{i-1}(t) - x_i(t) - L_{i-1}}{v_i(t) - v_{i-1}(t)} & v_i(t) > v_{i-1}(t) \\ \infty & v_i(t) \leq v_{i-1}(t) \end{cases} \quad (1)$$

Where $x_i(t)$ is the position of targeted (following) vehicle (i) at timestamp t , $L_{i-1}(t)$ is the length of preceding(leading) vehicle (i-1), $v_i(t)$ is the velocity of targeted (following) vehicle (i) at timestamp t .

As minimal TTC is an instantaneous variable, Time Exposed TTC (TET) and Time-Integrated TTC (TIT) were derived from TTC to evaluate the duration and integration of TTC smaller than a certain threshold, as in Figure 1b [15]. TET and TIT are calculated by.

$$TET_i = \sum_{t=0}^N \delta_i(t) \cdot \tau_{sc} \quad (2)$$

$$\delta_i(t) = \begin{cases} 1 & \forall 0 \leq TTC_i(t) \leq TTC^* \\ 0 & \text{othersize} \end{cases} \quad (3)$$

$$TIT_i = \sum_{t=0}^N [TTC^* - TTC_i(t)] \cdot \tau_{sc} \quad (4)$$

$$\forall 0 \leq TTC_i(t) \leq TTC^* \quad (5)$$

Where $\delta_i(t)$ is a switching variable between 1(indicates a signal of risk condition) and 0, τ_{sc} is the observation time interval (e.g.0.1s), TTC^* is the threshold of TTC value (e.g. 3s).

III. EXPERIMENTAL DESIGN

A. Simulation Platform

In this study, MATLAB Automated Driving System Toolbox™ provides virtual autonomous vehicle and autonomous driving algorithms for testing ACC with RSS. The tested vehicle is equipped with eight surround cameras which provide 360 degrees of visibility and 250m of forward range. A forward-facing radar sensor is also added to provide additional data, as shown in Figure 2.

ACC with sensor fusion algorithm, as shown in Figure 3, adjusts longitudinal acceleration rates of the subject vehicle

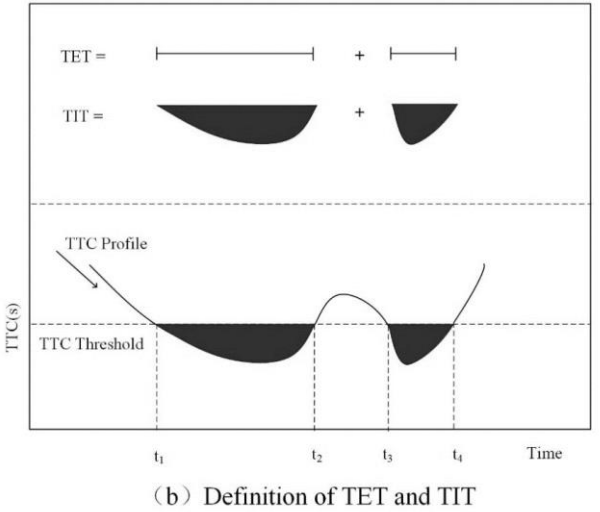
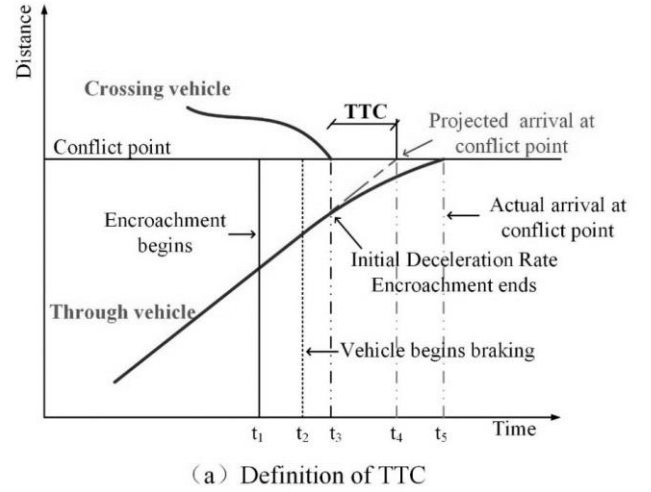


Figure 1. Graphical representation of TTC, TET and TIT

according to velocities of both subject and lead vehicles to maintain a safety front distance using model predictive control (MPC) [16,17]. The safety front distance is defined as

$$D_{safe} = D_{default} + T_{gap} \times V_x \quad (6)$$

where the default spacing $D_{default}$, and time gap T are design parameters and V_x is the longitudinal velocity of the subject vehicle. Considering the movement ability of the ego vehicle, the longitudinal acceleration is constrained to the range $[-3, 2] \text{ m/s}^2$. Moreover, in the MPC-based ACC algorithm, an optimization problem with a constraint that ensures front distance is always greater than the safety front distance (equation 7). The Vehicle Dynamics subsystem models the vehicle dynamics with the Bicycle Model.

$$\begin{aligned} & \text{minimize } |V - V_{set}|^2 \\ & \text{subject to } D_{relative} - D_{safety} \geq 0 \end{aligned} \quad (7)$$

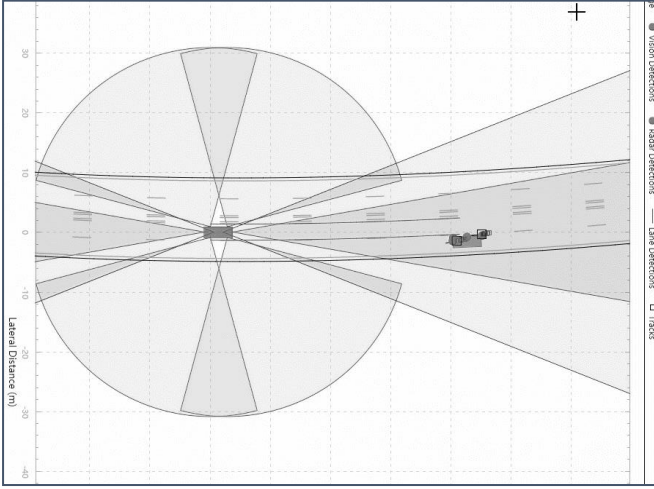


Figure 2. Setup of Multiple Sensors

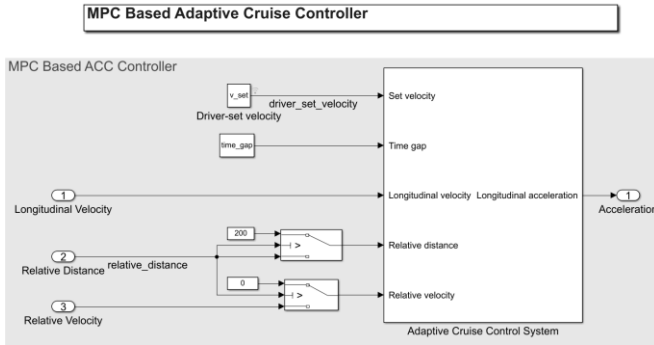


Figure 3. Adaptive Cruise Control Algorithm in Matlab Simulink

B. Embedding RSS in ACC

In the RSS model, the safe longitudinal distance on autonomous car-following maneuvers, is defined as the combination of distance travelled before braking response and breaking distance, as shown in Figure 4.

$$d_{\min} = \left[v_r \rho + \frac{1}{2} a_{\max, \text{accel}} \rho^2 + \frac{(v_r + \rho a_{\max, \text{accel}})^2}{2a_{\min, \text{brake}}} - \frac{v_f^2}{2a_{\max, \text{brake}}} \right]_+ \quad (8)$$

Where ρ denotes a response time, v_r , $a_{\max, \text{accel}}$ and $a_{\min, \text{brake}}$ denote the velocity, the maximum acceleration and the minimum deceleration of a rear car respectively, v_f and $a_{\max, \text{brake}}$ denote the velocity and the maximum deceleration of a front car. $[x]_+ = \max\{x, 0\}$

In this study, the RSS model is embedded in ACC algorithm according to the following logic: At each simulation step, the current front distance between subject vehicle and lead vehicle is computed. RSS safety distance is calculated at the same time. If the current distance is smaller than the minimum safe one, the vehicle decelerates with its maximum

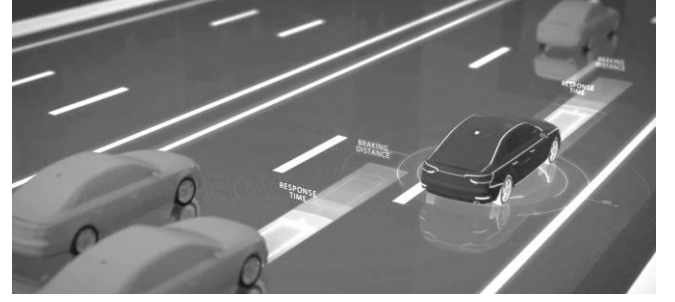


Figure 4. Definition of safety longitudinal distance in RSS model [15]

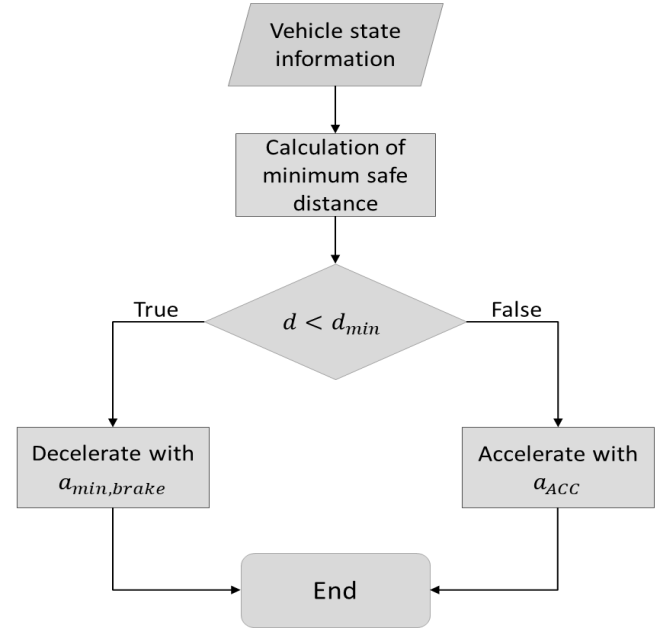


Figure 5. Incorporating RSS in Autonomous Driving Algorithms

deceleration. Otherwise, vehicle movement will be updated according to ACC, as shown in the Figure 5.

C. Simulation Scenarios and Algorithm Parameters

Typical car-following scenarios are designed according NHTSA pre-crash scenarios, as in Table 1 with front vehicle decelerates at the start of simulation with constant deceleration rate (-2 m/s^2). Performance of RSS model is highly affected by parameters in both ACC algorithm and RSS model. Reaction time of ACC, especially in commercial vehicle is found to be slightly smaller than human drivers. Values from 0.5~1.0s are applied in previous studies. Maximum and minimum deceleration rates in both RSS and ACC algorithms are dependent. Minimum deceleration rate cannot be too small as the controller may not be functional, maximum deceleration cannot be too high as safety distance will be too large and the result may become unrealistic.

To select critical parameters of ACC and RSS, this paper refers to the parameters of the Gipps' car-following model calibrated for Shanghai drivers [19]. The acceleration range of ACC is set to $[-3, 2.5] \text{ m/s}^2$. As car-following scenarios are designed to emergency deceleration scenarios of front vehicle, RSS minimum brake rate is set to the maximum observed deceleration of following vehicle in Gipps model, which is

approximately 3m/s^2 . The maximum deceleration of the RSS model is set to the maximum physically possible values of a vehicle, which is 8 m/s^2 . Table I lists selected parameters for simulation experiment. Results in Section 4 suggest the selected parameters are able to simulate realistic vehicle movements and ensure safety performance of the RSS model.

TABLE I. SIMULATION SCENARIOS AND ALGORITHM PARAMETERS

Time of deceleration	0s
Deceleration rate of front vehicle	-2 m/s^2
Velocity of front vehicle	$[50,40,30]\text{ km/h}$
Initial distance	$[80,70,60,50,40]\text{ m}$
Initial (cruise) velocity	Same as front vehicle
ACC reaction time	0.5 s
ACC default space	3.5 m
ACC time gap	1.5 s
RSS reaction time	1 s
RSS maximum brake rate	-8 m/s^2
RSS minimum brake rate	-3 m/s^2

IV. SAFETY EVALUATION OF RSS

A. Simulated Vehicle Movements

According to simulation parameters listed in Table I, there are 15 combinations of initial distances and initial velocities. Vehicle movements are simulated for each combination, Figures 6 and 7 shows simulated vehicle movements of two typical scenarios. Figure 6 is vehicle movements of a low risk scenario with initial distance of 70m, and initial velocity of 30km/h. Figures 6a and 6c plots relative distance and safety distances computed by RSS and ACC at each simulated time step without and with RSS. Figures 6b and 6d computes velocity of subject vehicle and TTC values at each simulate time step without and with RSS. RSS safety distances are only plotted for comparison in Figure 6a, they are not activated and will not affect vehicle movements. Moreover, as both vehicles have same initial velocity, initial TTC should be ∞ . For computation efficiency, the TTC values are capped at 10s.

According to Figures 6a and 6c, in the first half of simulation scenario, RSS safety distance is slightly larger than ACC distance. In both simulation scenarios with and without RSS embedded, the subject vehicle started deceleration 7s after simulation begins. However, after the relative distance continuously become closer, RSS distance becomes larger than relative distance and RSS model starts to control vehicle movements. Although safety distances computed by RSS and ACC are very close to each other after deceleration started, results in Figures 6b and 6d shows that after RSS is embedded, the vehicle velocity drops quicker during 8s-10s. Severity of vehicle conflicts is also reduced after embedding RSS as minimal TTC in Figure 6b is much lower than in Figure 6d.

Figure 7 shows a more severe scenario with initial distance of 80m, and initial velocity of 50km/h. In this scenario RSS safety distance becomes larger than relative distance about 4s after the simulation scenario starts. In contrary, ACC safety distance is much smaller compared to RSS throughout simulation process. Therefore, when ACC is not embedded with RSS, the subject vehicle started to decelerate after 6s. A crash is simulated as TTC reaches 0s at 10s in Figure 7b. However, after RSS is embedded, the vehicle started

deceleration at 3s to avoid collision. The conflict severity is significantly reduced as minimal TTC becomes 2s.

According to Figures 6 and 7, RSS is effective in both scenarios. However, the safety improvement is more obvious in Figure 7 with higher collision risk. As the goal of ACC is to maintain optimized velocity, it may lead to late reaction of autonomous vehicle under extreme conditions. The RSS

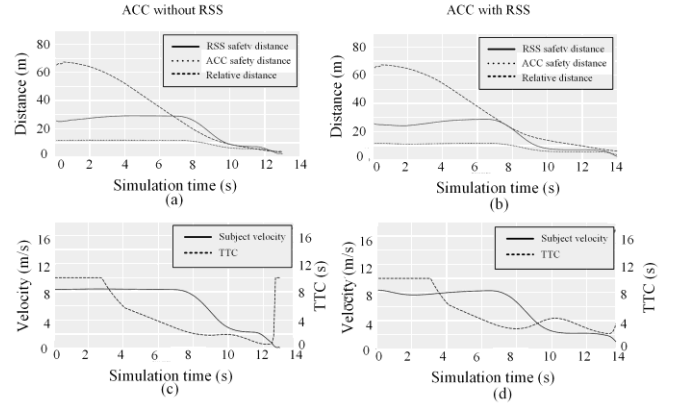


Figure 6. Simulated vehicle movements Initial (Distance=70m, Initial Velocity=30km/h)

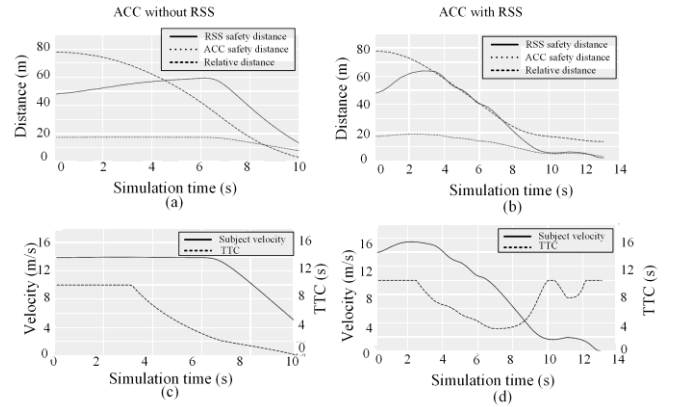


Figure 7. Simulated vehicle movements Initial (Distance=80m, Initial Velocity=50km/h)

model, in contrary, considers system reaction in high risk conditions and enhances the vehicle's early response ability. To further evaluate safety performances of RSS in ACC algorithms, all 15 scenarios are simulated and compared in Section 4.2.

B. SSM based Safety Evaluation

Figure 8 summarizes SSM computed for safety evaluation of each simulated car-following condition. TTC of 3s is selected for computation of TET and TIT, as shown in Figure 2b. Darker shadings in Figure 8 represent lower safety levels of each simulated condition. The thresholds are selected after evaluating the distribution of SSM, for high safety level is TET smaller than 1.5s, TIT smaller than 0.5s or minimal TTC larger than 3s. In contrary, if TET is larger than 3s, TIT is larger than 3s or minimal TTC smaller than 0.5s, the condition is considered as low safety level.

Safety effect of RSS can be observed from comparing SSM computed before and after RSS is embedded in ACC:

- According to Figures 8a 8c and 8e, safety level reduces at simulated conditions with large front gaps and high relative velocities. Collisions are observed at 4 conditions (relative velocity at 50km/h, and front gap at 50, 60, 70, 80). As the ACC algorithm in this study takes time gap as critical parameter, it is not able to identify high risk conditions with both high velocity and relative front gap. Another reason of low safety performance at large front gap condition is the ACC algorithm is based on optimization. The algorithm optimizes vehicle movement by keeping the most possible desired velocity and keep a relative time gap. Therefore, when the front vehicle suddenly decelerates with large front gap, ACC may response too late.
- Comparison shows the RSS model is able to improve safety performances at most car-following conditions. In particular, the effect of RSS is more obvious at high risk conditions with high velocity and large front gaps. As RSS is derived from driver's responses of acceleration rates, the model is more sensitive to deceleration of front vehicle than time gap based ACC algorithm.
- Moreover, according to Figure 8b, after ACC is embedded with RSS, in several conditions, minimal TTC values are still below 3s. More obvious safety improvement effects are observed from TET and TIT values. This indicate that RSS is more effective to reduce the time duration of conflict rather than on severity of conflict.

V. DISCUSSIONS AND CONCLUSIONS

A. Implication of RSS

This study evaluates safety impact of RSS on car-following maneuvers. ACC algorithm, which is a fundamental longitudinal vehicle control algorithm, is tested with and without RSS. Results show RSS is effective to improve safety performance of ACC. As ACC is mainly based on time gap, when front gap is larger than ACC safety distance, deceleration response is too late to reduce collision risk. Such shortcomings of low level autonomous driving algorithms can be improved by embedding with RSS.

Findings of this study, through comparing safety performance with and without RSS, can be further applied to identify proper scenarios and to improve the existing RSS approach. Firstly, as RSS is more effective on situations with high following velocities and large front gap, it is recommended to embed RSS at such conditions. Activating RSS on certain other situations may enlarge safety distance and can be "overly conservative". Furthermore, simulation results suggest RSS has more obvious time dependent effect. To reduce its "overly conservative" problem, RSS needs to be further improved.

B. Limitations and Future Works

One limitation of this study is that parameters are selected according to existing studies. Although simulation results show that the selected parameter sets are effective and not too conservative for the tested simulation scenarios, in future studies, more combinations of RSS parameters should be

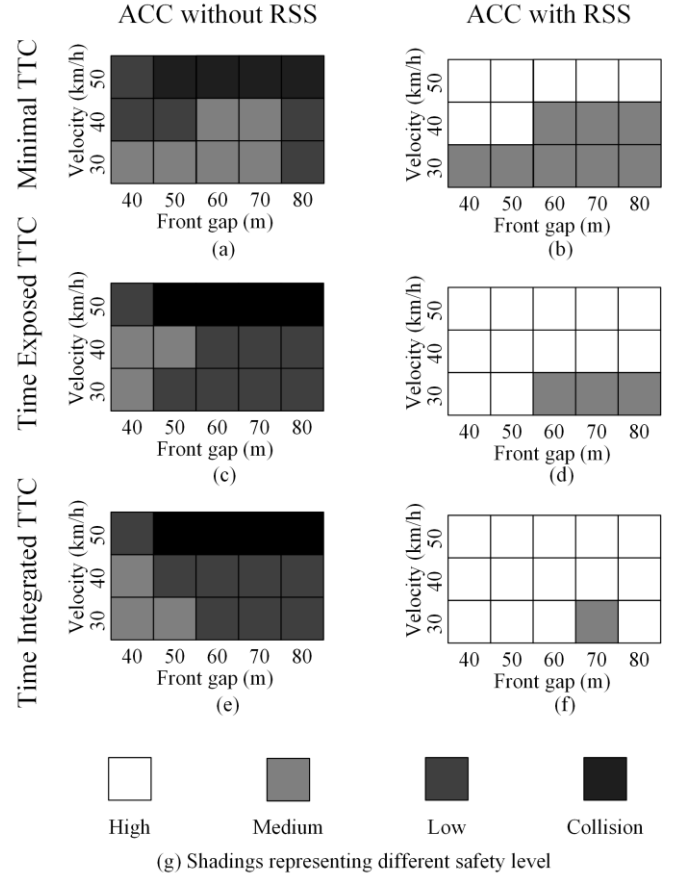


Figure 8. Safety performance of RSS according to SSM

tested and compared for various scenarios for model optimization. Moreover, as an algorithm independent safety guarantee model, RSS can be applied to evaluate safety performance of multiple autonomous driving algorithms.

REFERENCES

- [1] Autonomous Vehicle Disengagement Reports 2018. California's Department of Motor Vehicles (DMV). https://www.dmv.ca.gov/portal/dmv/detail/vr/autonomous/disengagement_report_2018. [Retrieved on 4th April 2019]
- [2] Implementing the RSS Model on NHTSA Pre-Crash Scenarios. Mobileye. https://www.mobileye.com/responsibility-sensitive-safety/rss_on_nhtsa.pdf [Retrieved on 4th April 2019]
- [3] Mattas K., M. Makridis, M. A. Raposo, B. Ciuffo, "How the Responsibility-sensitive Safety Framework Affects Traffic Flows on a Freeway Microsimulation Scenario," presented at the 98th Transportation Research Board Annual Meeting, Washington, D.C., January 13–17, 2019.
- [4] Rieger G., Scheef J., Becker H., Stanzel M., Zobel R., "Active Safety Systems Change Accident Environment of Vehicles Significantly-A Challenge for Vehicle Design." In Proceedings of the International Conference on Enhanced Safety of Vehicles, 2005.

- [5] Magdici S., Althoff M., "Adaptive Cruise Control with Safety Guarantees for Autonomous Vehicles." In Proceedings of the International Federation of Automatic Control (IFAC), pp. 5774-5781.
- [6] Alipour-Fanid A., Dabaghchian M., Zhang H., and Zeng K. "String stability analysis of cooperative adaptive cruise control under jamming attacks." 2017 IEEE 18th International Symposium on High Assurance Systems Engineering (HASE), pp. 157–162, 2017.
- [7] Milanés V., Shladover S. E., Spring J., Nowakowski C., Kawazoe H., and Nakamura M. "Cooperative adaptive cruise control in real traffic situations." IEEE Transactions on Intelligent Transportation Systems, vol. 15, no.1, pp. 296–305, 2014.
- [8] Takahama T., Akasaka, D., "Model Predictive Control Approach to Design Practical Adaptive Cruise Control for Traffic Jam" International Journal of Automotive Engineering, vol. 9, no. 3, pp. 99-104, 2018.
- [9] Najm W. G., Smith J. D. and Yanagisawa M., "Pre-crash Scenario Typology for Crash Avoidance research," DOT HS, 2007.
- [10] Hydén C., "The Development of a Method for Traffic Safety Evaluation: The Swedish Traffic Conflicts Technique," Lund University, 1987.
- [11] Oh C., and Kim T., "Estimation of Rear-End Crash Potential Using Vehicle Trajectory Data," Accident Analysis and Prevention, vol. 42, no. 6, pp. 1888–1893, 2010.
- [12] Zheng L., Ismail K. and Meng X., "Traffic Conflict Techniques for Road Safe Open Questions and Some Insights," Canadian Journal of Civil Engineering, vol. 41, no. 7, pp. 633–641, 2014.
- [13] Shi X., Wong Y. D., Li M. Z. and Chai C., "Key Risk Indicators for Accident Assessment Conditioned on Pre-Crash Vehicle Trajectory," Accident Analysis and Prevention, vol. 117, pp. 346–356, 2018.f
- [14] Minderhoud M. M. and Bovy P. H., "Extended Time-To-Collision Measures for Road Traffic Safety Assessment," Accident Analysis and Prevention, vol. 33, no. 1, pp. 89–97, 2001.
- [15] van der Horst. "A time-based analysis of road user behavior in normal and critical encounters." Doctoral Dissertation. Delft University of Technology, No. HS-041255, 1990.
- [16] van der Heijden R., Lukaseder T., and Kargl F. "Analyzing attacks on cooperative adaptive cruise control (CACC)." IEEE Vehicular Networking Conference (VNC), pp. 45–52, 2017.
- [17] Ploeg J., Semsar-Kazerooni E., Lijster G., van deWouw N., and Nijmeijer H. "Graceful degradation of CACC performance subject to unreliable wireless communication." 16th International IEEE Conference on Intelligent Transportation Systems (ITSC), pp. 1210–1216, 2013.
- [18] Li Y., Xu C., Xing L., and Wang W. "Integrated cooperative adaptive cruise and variable speed limit controls for reducing rear-end collision risks near freeway bottlenecks based on micro-simulations." IEEE Transactions on Intelligent Transportation Systems, vol. 18, no. 11, pp. 3157–3167, 2017.
- [19] Zhu M. X., Wang X. S., Tarko A., Fang S. E. "Modeling car-following behavior on urban expressways in Shanghai: A naturalistic driving study." Transportation Research Part C: Emerging Technologies, vol. 93, pp. 425–445, 2018.